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NEUTRON STAR MATTER AT SUB-NUCLEAR DENSITIES

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Abstract: An extremely simple form for the energy density of a nuclear many-body system derived from the two-body nucleon-nucleon interaction is used to determine the ground state configuration of matter at sub-nuclear density As the baryon density is increased, nuclei become progressively more neutron rich until neutrons eventually escape, yielding a Coulomb lattice of bound neutron and proton clusters surrounded by a dilute neutron gas The clusters enlarge and the lattice constant decreases with increasing density, approaching a completely uniform state near nuclear density

1. Introduction

There exists sufficient astrophysical evidence that pulsars are in fact neutron stars that it is worthwhile to seriously investigate the equation of state of dense baryonic matter utilizing one's current understanding of theoretical nuclear physics. In this present work, consideration is restricted to that limited regime in which the two tacit assumptions of nuclear physics are satisfied, namely that the temperature be low compared with characteristic nuclear excitation energies and that the meson degrees of freedom may be legitimately suppressed and replaced by a phenomenological two-body potential determined by scattering data and deuteron properties.

The first condition appears to be satisfied for observed pulsars, since the surface temperature may be inferred to be less than 10^8 K, corresponding to 10 keV, and the thermal conductivity is expected to be very high. Thus, ignoring the early stages of formation, on the relevant temperature scale of nuclear energies, neutron stars are exceedingly cold and may be treated in first approximation as being at a temperature of absolute zero. Hence it is legitimate to use the same zero-temperature perturbation expansions which are used in nuclear many-body theory, and the equation of state is completely specified by calculating the binding energy per baryon as a function of baryon density.

The suppression of meson degrees of freedom in nuclear physics has been extremely successful. At a density of 0.17 nucleons/fm^3, corresponding to the interior density of a heavy nucleus, the average distance between nuclei is 2 2 fm, whereas

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the one-pion exchange force has a range of 1.4 fm. Thus, it appears quite reasonable to first approximate the interaction between two nucleons in nuclear matter as the interaction which would occur in free space, and then treat as a small perturbation the explicit meson many-body effects. The lowest-order example of such an effect would be the three-body force arising from the interaction of a pion in transit between two nucleons with yet a third nucleon Such processes have been investigated in nuclear matter and yield a correction to the binding energy per particle on the order of 2 MeV out of a total potential energy contribution of 40 MeV [ref. [1])]. Thus, present evidence indicates that if we restrict our attention to nuclear configurations with densities less than or equal to that occurring in the interior of large nuclei, the structure should be dominated by the two-body nucleon-nucleon interaction and explicit meson many-body effects should produce only very small corrections amenable to perturbation theory

Having determined the upper limit on densities to be considered to be that occurring in the interior of large nuclei, hereafter referred to simply as nuclear density and having the value 0.17 nucleons/fm^3 = 2.8×10^{14} g/cm^3, we now consider the lower limit.

At densities above 10^7 g/cm^3 for temperatures below 10^8 K, matter is expected to be a solid. This is because the Coulomb interaction between ions is only weakly screened and at sufficiently low temperatures, the Coulomb energy is minimized by a bcc lattice. Applying the Lindemann melting criterion which requires the mean fluctuations of ions to be small compared with the average ion spacing, the ions are expected to form a solid below the temperature $T_m \approx Z^2e^2/(100r_Z)$, where r_Z is the average spacing between ions [2]). For ^{56}Fe, this solidification occurs by the time one reaches a density of 10^7 g/cm^3, and as the density increases, r_Z decreases, assuring that the condition will be maintained at all subsequent densities. At extremely high densities, one might worry about the Coulomb lattice breaking up solely due to the zero point motion of the ions, but for the highest-density configuration treated in this work, $_{32}$Ge at 1.4×10^{14} g/cm^3, the zero point fluctuation is still very small compared with the ion spacing.

For Coulomb lattices of ions at densities between 10^7 and 10^9 g/cm^3, the equation of state is determined directly from the experimentally observed mass table using straightforward corrections for the electrons and lattice Coulomb energy [3]). At $\rho = 6.2 \times 10^9$ g/cm^3, the Fermi energy of the relativistic electrons shifts the energy balance favoring neutron rich nuclei so strongly that ^{84}Se, which is unstable but experimentally observable, gives way to ^{82}Ge which has not been observed but may be reliable extrapolated [4]) Extrapolations from the mass table continue to be reliable up to a density of roughly 4.3×10^{11} g/cm^3, at which point the neutrons in ^{118}Kr are just barely bound and any further increase in density causes the last few neutrons to "drip" out of the nuclei and form a low-density neutron gas in the intervening space between the nuclear clusters. Although the theory presented in this present work should also be valid in the entire pre-drip sequence of extrapolated nuclei, we believe the mass

table extrapolations are satisfactory well below the drip point and have concentrated our attention on the last sequence of pre-drip nuclei with 82 neutrons (^{124}Mo, ^{122}Zr, ^{120}Sr and ^{118}Kr), and on the free neutron regime after drip occurs, between 4×10^{11} and 2×10^{14} g/cm^3.

In this work we shall not be concerned directly with the stellar structure of neutron stars, but rather, only with the equation of state which specifies the pressure which must be used in the general relativistic equation of hydrostatic balance However, the equation of state derived in this work turns out to be extremely close to that used in numerical solutions of the Tolman-Oppenheimer-Volkoff equation by Baym, Pethick and Sutherland [5]) so the general features of their results for light neutron stars should also pertain to this present work. From their results, it is observed that the very lightest neutron stars do not exceed nuclear density and that the region from 4×10^{11} to 2×10^{14} g/cm^3 is the most crucial in determining the structure and stability of these stars. In particular, one may observe qualitatively the effect of neutron drip and the approach to nuclear density on the equation of state At 4×10^{11} g/cm^3, where neutron drip occurs, the sudden increase in slope of the stellar density distribution indicates the appreciable softening of the equation of state when neutrons begin to drip out of nuclei. Similarly the flatness of the density distribution near nuclear density reflects the stiffness of the equation of state once nuclei begin to touch and the gravitational pressure attempts to compress the nucleons to a density higher than that occurring in a free nucleus. It is the need for quantitative precision in describing both the softness at the neutron drip point and stiffness near nuclear density that motivated the present application of a microscopic theory of nuclear structure to examine in somewhat greater detail the equation of state in this regime.

2. Previous investigations

It is natural that the first attempts to treat the free neutron regime should be based on extrapolations from the semi-empirical mass formula [6, 7]). The limitations of such an approach arise, obviously, from the fact that the mass formula parameters are determined only by a very restricted region of nuclear configurations: the ratio of protons to neutrons is greater than ≈ 0.6, the chemical potential (minus the removal energy) for neutrons and protons is on the order of -8 MeV and there is zero external pressure on the nucleus. In the free neutron regime, however, the configurations are very far from satisfying these conditions. The ratio of protons to neutrons in nuclei is 0.1 to 0.3; the neutron chemical potential in the nucleus must equal that of the free gas and thus cover a range from 0 to $+20$ MeV; the pressure from the exterior neutron gas becomes significant, and the surface becomes much more diffuse and instead of approaching zero density outside the nucleus, approaches the neutron gas density For all these reasons, extrapolations based on the semi-empirical mass formula are extremely unreliable, and one is forced to undertake a more fundamental theory.

Baym, Bethe and Pethick [5]) (hereafter denoted BBP) introduced the information which is unobtainable from the mass formula by means of the theory of uniform nuclear matter. Based on nuclear matter calculations with the Reid [8]) soft core potential by Siemens [9]) in the region of roughly equal neutron and proton densities and by Siemens and Pandharipande [10]) in the region of almost pure neutron matter, BPP obtained a simple parameterization of the nuclear binding energy per nucleon surface as a function of density and ratio of protons to neutrons. Both Coulomb and gravitational interactions are necessarily omitted from these infinite nuclear matter calculations. In addition, since only two-body reaction matrix diagrams are included, a phenomenological correction was applied to include the effects of all higher-order diagrams, three-body forces, and relativistic corrections. Since such corrections enter most strongly in the $^3S_1$–$^3D_1$ coupled partial waves which are connected by the tensor force, the phenomenological correction was defined to occur only in isospin $T = 0$ states and thus acts only between unlike particles By means of this correction, the minimum of the saturation curve was shifted from 11 MeV to 16.5 MeV at a density corresponding to $k_F = 1.43$ fm^{-1}

Having determined the binding energy surface for all densities and ratios of protons to neutrons, BBP treat the Coulomb lattice of nuclei surrounded by a neutron gas in the Wigner-Seitz approximation [11]). The unit cell is divided into a uniform-density interior nucleus comprised of neutrons at high density and protons at low density, and a uniform neutron gas exterior, comprised of neutrons at low density. A phenomenological surface energy expression motivated by Thomas-Fermi theory corrects for the presence of a finite surface and the Coulomb energy is calculated assuming a constant density of electrons filling the entire cell to yield charge neutrality. Then at a specified baryon density, the ground state configuration is determined by requiring that the system be stable against β-decay, that the pressure inside the nucleus equal the pressure in the outside gas, and that the neutron chemical potential inside the nucleus equal that of the outside gas, and by minimizing the total binding energy with respect to the radius of the unit cell.

Whereas this approach constitutes a clear improvement over previous work, it is still subject to two limitations. The first is an incomplete theory of the nuclear surface, with the general form being specified by Thomas-Fermi theory, but with the actual surface thickness being determined from the neutron Fermi wavelength instead of variationally. The second is the complete absence of nuclear shell effects, which are already observed to play an important role in determining the composition of the ground state matter prior to neutron drip.

Similar calculations by Arponen [12]) and Barkat, Buchler, and Ingber [13]) utilize slightly different parameterizations of the nuclear energy surface and determine ground state density distributions variationally in a Thomas-Fermi theory. The surface energy is included by introducing a gradient term in the expression for the energy as a functional of the density with a phenomenological coefficient adjusted to fit ordinary nuclei.

The semi-classical Thomas-Fermi approximation has been compared with the quantum mechanical Hartree-Fock method for a semi-infinite surface by Ravenhall *et al.* [14]). Using the Skyrme interaction, they demonstrate significant differences between the density distributions and equilibrium configurations with the two methods. The final result they obtain in Hartree-Fock theory for the charge of the nucleus as a function of matter density lies between the results of BBP and Barkat, Buchler and Ingber and constitutes the most reliable calculation discussed thus far. We will subsequently compare the calculations of this present work with those of Ravenhall *et al.* and show how the differences arise from curvature terms in the surface energy and nuclear shell effects, both of which are necessarily omitted in a semi-infinite matter calculation

3. Theory of finite nuclei

Present evidence indicates that a strongly repulsive core in the nucleon-nucleon interaction plays an essential role in producing finite nuclei which obtain the proper binding energy without collapsing to unphysically high densities. Once one adopts a potential with a strongly repulsive core, such as the Reid soft core potential used in this work, perturbation theory can no longer be ordered simply in terms of numbers of interactions of the bare potential, but rather must be re-expressed in terms of the reaction matrix.

The reaction matrix G sums all orders of ladder diagrams containing the bare interaction and unoccupied intermediate states. Thus, physically, it permits two interacting particles to interact any number of times when they approach each other, thereby allowing the two-body wave functions to respond to the presence of the potential by generating correlations which strongly decrease the probability of the particles penetrating into the repulsive core region. All the physical information concerning the reaction matrix is contained in the correlated wave function, and since the Pauli principle prohibits scattering to any final states other than the original states, the correlated wave function heals rapidly to the uncorrelated wave function beyond the range of the component of the two-body force generating the correlations.

The simplest system in which to investigate the reaction matrix is uniform nuclear matter, and nuclear matter theory is described in great detail by two recent extensive review articles by Bethe [16]) and Sprung [17]). Since analogous calculations in large finite nuclei are presently impractical, the information obtained by investigating two-body correlations occurring between nucleons in nuclear matter is applied to finite nuclei via the local density approximation (LDA) Physically, one argues that the short-range correlations between two nucleons in the interior of ^{208}Pb, which is roughly 13 fm in diameter, should be the same as for two nucleons in infinite uniform nuclear matter at the same neutron and proton density. For nucleons near the nuclear surface, which is 2 fm thick, the argument is still quite good for the 0.4 fm correlations due to the repulsive core, and begins to break down only for the longer-range tensor force correlations.

Calculations of ground state properties of spherical closed-shell nuclei using the LDA are described in detail in refs. [18-20]). Essentially, an energy- and density-dependent effective interaction is constructed from the nuclear matter reaction matrix, and from second-order perturbation theory, it follows that both the energy and density dependence contribute to saturation by making the interaction less attractive at high densities.

Once one has accepted an effective interaction which is density and energy dependent, it is straightforward to write out the expression for the energy of a Slater determinant and to functionally differentiate with respect to each single-particle wave function. In addition to the usual Hartree-Fock terms obtained from varying $\Psi_m^*(r_1)$ and $\Psi_n^*(r_2)$ appearing explicitly in

$$\sum_{mn}\int\int \psi_m^*(r_1)\psi_n^*(r_2)G(|r_1-r_2|, \rho(\tfrac{1}{2}(r_1+r_2)), e_m+e_n) \times [\psi_m(r_1)\psi_n(r_2)-\psi_n(r_2)\psi_m(r_1)]\mathrm{d}^3r_1\,\mathrm{d}^3r_2\,,$$

one also obtains terms of the form $(\partial G/\partial\rho)(\partial\rho/\partial\psi_s^*)$ and $(\partial G)/\partial e_m)(\partial e_m/\partial\psi_s^*)$. These terms express the fact that by increasing the probability of finding particle s at some region in the nucleus, the total energy contribution from that region is not only increased because of the pair interactions of particle s with all other nucleons, but it is also slightly decreased by the fact that two other particles, say m and n interacting in that region, now find their interactions less attractive due to the fact that particle s has now made some phase space unavailable for intermediate scattering states and has slightly increased their energy denominators. These extra terms in the resulting density-dependent Hartree-Fock (DDHF) equations, loosely referred to as "rearrangement terms", are crucial to obtaining saturation in finite nuclei and in obtaining agreement with experimental single-particle energies and binding energies.

The effective interaction described above is actually defined in detail to reproduce exactly the same nuclear matter saturation curve as Siemens' original G-matrix [9]) when evaluated in a Fermi gas of plane wave states. Thus, in order to obtain the proper binding energy per particle, it is necessary to introduce a phenomenological correction to include all the higher-order terms in nuclear matter theory. In the same spirit as BBP, this was chosen to be a short-range force acting only between unlike particles and for the results presented in this work, the two parameters were adjusted to give 16.53 MeV binding energy per particle at a saturation density of $k_F = 1.33$. It should be emphasized that little ambiguity is introduced into the theory, since theoretically, higher-order corrections are much smaller for a pure neutron gas than for nuclear matter, due to the absence of the ^{3}S–^{3}D tensor force, and the adjustment for nuclear matter is constrained by the properties of finite nuclei.

The results of DDHF theory presented in refs. [18,19]) yield extremely good agreement with experimentally observed bulk properties of finite nuclei. In particular, since we are especially interested in a reliable theory of the nuclear surface, it is

important to note that the accurate agreement with electron scattering results of density distributions throughout the periodic table is a stringent test of the exact balance of volume energy, surface energy, Coulomb energy and symmetry energy in the DDHF theory.

The DDHF theory as described above is, unfortunately, too computationally time-consuming to apply to the configurations of up to 5000 nucleons in a unit cell which we wish to investigate in the free neutron regime. Since the primary computational complication in the DDHF theory arises from the non-local exchange term which necessitates both the Legendre expansion of the exchange potential and the explicit solution of a non-local Schrödinger equation, it is useful to develop a systematic expansion of the exchange term which treats it in a much more convenient manner. The density matrix expansion (DME), which is described in detail in ref. [21]), expands the two-body density matrix $\rho(r_1, r_2) = \sum_n \psi_n^*(r_1)\psi_n(r_2)$ in a series in the relative coordinate $r_1 - r_2$ and appropriate derivatives of the wave functions at the center-of-mass point $R = \frac{1}{2}(r_1 + r_2)$. The first term is the Slater mixed density which is exact for uniform matter and a good first approximation even in the nuclear surface. The first correction term involves $\rho(R) = \sum_n |\psi_n(R)|^2$, the density, and

$$\tau(R) = \sum_n |\nabla\psi_n(R)|^2,$$

the kinetic energy density, and gives an extremely accurate approximation for the density matrix within the range of the nuclear force throughout the nucleus.

Truncation of the expansion of the density matrix at second order yields an extremely simple factorized expression of specified functions of the relative coordinate multiplying $\rho(R)$, derivatives of $\rho(R)$, and the kinetic energy density, $\tau(R)$. The total Hamiltonian density may then be obtained by performing the integral over relative coordinates of the density matrix squared times the effective interaction. Distinguishing between proton and neutron densities and kinetic energies, the final form of the Hamiltonian density is

$$H(R) = \frac{\hbar^2}{2M}(\tau_n + \tau_p) + A(\rho_n, \rho_p) + B(\rho_p, \rho_n)\tau_p + B(\rho_n, \rho_p)\tau_n$$
$$+ C(\rho_p, \rho_n)|\nabla\rho_p|^2 + C(\rho_n, \rho_p)|\nabla\rho_n|^2 + D(\rho_n, \rho_p)\nabla\rho_n \cdot \nabla\rho_p,$$

where the functions A, B, C, and D are specified integrals involving the effective interaction at the local neutron and proton density and spherical Bessel functions.

One useful conceptual feature of this expansion is the fact that $H(R)$ may be separated into a term which is precisely the nuclear matter potential energy per particle, the kinetic energy, and terms which depend only on integrals of the long-range part of v_{eff} times τ and gradients of ρ. The long-range part of v_{eff} is very close to the bare potential, which in turn is the most unambiguously determined part of the nuclear potential. Thus, the finite nucleus corrections are virtually independent of the model of the nuclear force, and all ambiguities concerning the short-range repulsion, off-

TABLE 1

Expansion coefficients for the symmetric function $V(\rho_\alpha, \rho_\beta) = \Sigma_{IJ} V_{IJ} \rho_\alpha{}^I \rho_\beta{}^J$ with redundant off-diagonal elements omitted

$V_{IJ}{}^{NM}$	0	1	2	3	4	5	6
0	0.0						
1	0.0	-1.29058×10^{3}					
2	-2.60473×10^{2}	3.48227×10^{3}	5.41286×10^{4}				
3	2.70730×10^{3}	2.08122×10^{4}	$-9\ 01970\times10^{5}$	1.68268×10^{7}			
4	$-2\ 97146\times10^{4}$	-9.24196×10^{4}	-1.69465×10^{6}	-2.96778×10^{7}	3.35473×10^{8}		
5	1.83110×10^{5}	8.17815×10^{5}	2.40883×10^{7}	-2.72504×10^{8}	-2.39553×10^{9}	4.55011×10^{10}	
6	-4.30956×10^{5}	-4.77095×10^{6}	-2.89827×10^{7}	2.15640×10^{9}	-1.62585×10^{10}	1.29625×10^{9}	-1.88806×10^{11}

energy-shell behavior, and higher-order corrections enter only through the nuclear matter energy surface as a function of ρ_n and ρ_p.

Technically, the DME is much simpler than the DDHF theory, involving only a local Schrödinger equation with a position-dependent effective mass. The results of this simplified theory have been compared in detail with the DDHF results [21]) and are in excellent agreement. Because of the computational simplicity of the DME, it was possible to search more extensively over the parameters of the higher-order correction adjustment and thereby obtain binding energies per particle which agree with experimental values within 0 1 MeV.

The only additional change required to apply this theory of finite nuclei to the free neutron regime was to introduce a more careful parameterization of $A(\rho_n, \rho_p)$ in the region of very low neutron density so as to accurately reproduce the uniform nuclear matter calculations of Siemens and Pandharipande [10]).

In the finite nucleus case, $V^{NM}(\rho_n, \rho_p)$, which enters into the construction of $A(\rho_n, \rho_p)$, was approximated by a polynomial of the form·

$$V(\rho_n, \rho_p) = \sum_{I,J=0}^{7} V_{IJ} \rho_n^I \rho_p^J .$$

The coefficients V_{IJ} were determined by a least-squares fit to values of $V(\rho_n, \rho_p)$ calculated on a two-dimensional 16×16 mesh of points evenly spaced in steps of $\Delta\rho = 0.01$ fm^{-3}. Since densities below 0.01 fm^{-3} play a negligible role in determining the structure of finite nuclei, there was no need to constrain the form of V to closely reproduce the Siemens-Pandharipande curve for low-density neutron matter. For the present application, however, in the free neutron regime, the binding energy per particle is dominated by the energy of the low-density neutron gas, so it was imperative that $V(\rho_n, \rho_p)$ was refitted, requiring $V_{00} = V_{10} = V_{01} = 0$, shifting the mesh such that the first step occurred at $\rho = 9 \times 10^{-4}$ fm^{-3}, and weighting the least-squares fit in the region of low density as well as at the nuclear matter saturation point.

The resulting values of V_{IJ} obtained from this fit are tabulated in table 1, and were used with the original values of B_{IJ}, C_{IJ}, and D_{IJ} from ref. [21]).

Although our primary emphasis is on the free neutron regime, the DME theory of finite nuclei has been used to check the extrapolation by Myers and Swiatecki [4]) of the last pre-drip sequence of nuclei, $^{122}_{40}$Zr, $^{120}_{38}$Sr, and $^{118}_{36}$Kr As shown in table 2,

TABLE 2

BE/A for pre-drip nuclei

Nucleus	Extrapolated value (MeV)	DME (MeV)
$^{122}_{40}$Zr	7.67	7.54
$^{120}_{38}$Sr	7.45	7.32
$^{118}_{36}$Kr	7.20	7 05

the DME results are in good agreement with the extrapolated values, and in particular, the differences between neighboring nuclei are in excellent agreement. Thus, we corroborate the results of Baym *et al.* [3]) for the composition of matter prior to neutron drip. The spatial distribution of protons in ^{118}Kr is shown in fig. 4 for subsequent reference.

4. The neutron drip regime

The equation of state in the neutron drip regime is obtained by minimizing the energy per nucleon of spherically symmetric configurations of nucleons in a Wigner-Seitz unit cell.

For charge neutrality, a cell with Z protons contains an equal number of electrons. Because of the large Fermi-Thomas screening length, $1/k_{\mathrm{FT}} = 4e^2/\pi\hbar c k_e$, where k_e is the electron Fermi wave number, the electrons are approximated by a uniform gas, and the small screening correction is neglected [5]). The kinetic energy per electron is then

$$\frac{T}{Z} = m_e c^2 \left\{ \frac{3}{8x^3} [x(1+2x^2)(1+x^2)^{\frac{1}{2}} - \ln (x + \{1+x^2\}^{\frac{1}{2}})] - 1 \right\},$$

where $x = \hbar k_e / m_e c$. The Coulomb exchange energy for the electrons is of the order of $e^2/\hbar c$ times the mean electron kinetic energy and is also neglected. With these approximations, the electron chemical potential is

$$\mu_e = (k_e^2\hbar^2c^2 + m_e^2c^4)^{\frac{1}{2}} - m_e c^2 + \hbar c \int \mathrm{d}^3 r_1 \, \mathrm{d}^3 r_2 (\rho_p(r_1) + \rho_e(r_1)) \frac{e^2}{|r_{12}|} \rho_e(r_2).$$

The nuclear energy for the unit cell is expressed in terms of the nucleon wave functions using the DME Hamiltonian density functional. The direct Coulomb energy is calculated straightforwardly from the electron and proton densities, and the Slater approximation is used for the nuclear exchange Coulomb energy. Since the total energy expression is variational in the nuclear wave functions, the proton and neutron chemical potentials, μ_p and μ_n, are given by the eigenvalues of the last occupied orbitals.

In addition to the approximations discussed previously in connection with the DME, it is extremely useful to omit the spin-orbit splittings for neutron states, thereby reducing the total number of wave functions to be considered by almost one half. Since the one-body spin-orbit potential is proportional to $(1/r)\mathrm{d}\rho/\mathrm{d}r$ and the spectrum of the continuum neutron states is essentially determined by the uniform neutron gas region, the effect of the spin-orbit force on the neutron level ordering is negligible Thus the density and energy density obtained by filling the barycentric states representing the average of the levels $j = l - \frac{1}{2}$ and $j = l + \frac{1}{2}$ with $2(2l+1)$ particles should be an excellent approximation. For protons, however, the spin-orbit splitting is crucial in determining the level ordering and thus which orbitals are

actually occupied. In order to retain a theory which is completely variational, the appropriate proton spin-orbit term to be added to the energy density [21]) is

$$H_{s\,o} = \tfrac{1}{2}W[\nabla(\rho_n+2\rho_p)\cdot \boldsymbol{J}_p],$$

where $\boldsymbol{J}_p$ is the proton spin density

$$\boldsymbol{J}_p = (\boldsymbol{r}/4\pi r^4)\sum_i [j_i(j_i+1)-l_i(l_i+1)-\tfrac{3}{4}]u_i^2(r)$$

and W is a constant determined from the two-body spin-orbit force

By construction of the DME Hamiltonian functional, pairing correlations have been entirely omitted from the theory. Since pairing is a non-negligible effect for a low-density neutron gas [22]), it is important at this stage to explain why it has been omitted from the present work and how the present results should be modified. It will be useful to consider separately the effect of pairing on the nucleus-like cluster at the center of the unit cell and on the energy of the low-density neutron gas filling the space between lattice sites.

The effect of pairing in a Hartree-Fock calculation of finite nuclei extremely similar to the DME theory described above has recently been investigated by Vautherin [23]), and is relevant to the structure of the cluster at the unit cell center. In particular, it is shown that unless one deals specifically with the pairing matrix elements of the true two-body interaction, there is an arbitrariness in the binding energy of nuclear states obtained from various pairing Hamiltonians which yield identically the same nuclear wave functions. Including all the relevant pairing matrix elements of the effective interaction would be prohibitive, and since pairing energies in nuclei are at most on the order of 0.1 MeV per particle, the effect on the equilibrium configuration and equation of state would be negligible. Hence it was decided to avoid the arbitrariness of introducing a phenomenological pairing Hamiltonian altogether, since, in contrast to deformed calculations, it played no essential role.

For a low-density uniform neutron gas, however, pairing gives a significant increase in the binding energy per particle [22]). Ideally, it would be desirable to include this energy in the nuclear matter functional $V^{NM}(\rho_n, \rho_p)$, and then correct the final result for the degree to which the lattice of nucleus-like clusters inhibits the pairing which would occur in a completely uniform medium. Since adequate BCS calculations with the Reid potential were not available for this work, we have chosen to use the results of Siemens and Pandharipande [10]), which omit pairing, and to indicate how the present theory may be trivially corrected when the relevant pairing results are available. Throughout the low-density end of the free neutron regime, the nuclear clusters occurring in the center of a unit cell are insensitive to small changes in the energy functional for the neutron gas. Hence, the composition and energy of the equilibrium clusters will be essentially unchanged by the inclusion of pairing. Since most of the baryons in the cell are neutrons in the gas, their energy change may be computed directly from the change in the neutron gas energy. Thus, the

modified equation of state may be obtained straightforwardly from the data in table 3 and the BCS pairing energy, without repeating the lengthy self-consistent searches reported in this work.

The final approximation required to completely specify the theory concerns the boundary conditions on the neutron wave functions at the edge of the Wigner-Seitz unit cell. The neutron gas between lattice sites is essentially uniform, and the com-

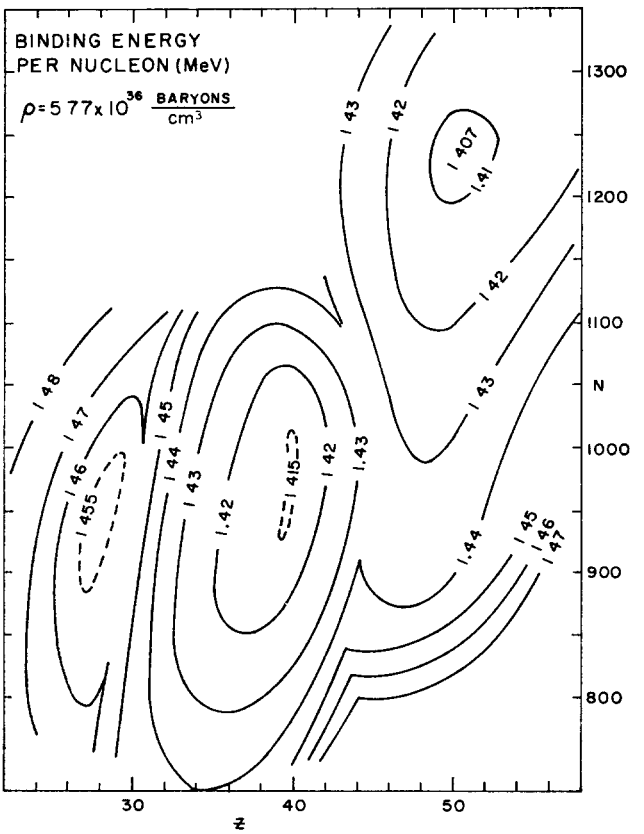


Fig. 1 Contour plot of energy per particle as a function of number of protons and neutrons in a unit cell.

plicated close-packed polyhedron is replaced by a spherical cell of the same volume in the usual way. Since the neutrons are essentially described by a statistical theory, one is free to choose any convenient boundary condition which is consistent with a statistical density distribution at the cell edge It turned out to be convenient to require that at the cell radius, all wave functions of even parity vanish and the derivatives of odd-parity wave functions vanish. This choice has the advantage of making the density approach roughly the average density at the cell edge, rather than zero or twice the average density, as would occur if either condition were applied separately

to all wave functions. In addition, to prevent the growth of any unphysical fluctuations in density near the cell edge, at each iteration, ρ and τ were replaced in the region within $1/k_F$ of the cell edge by their average values computed over the rest of the uniform gas region of the cell. These boundary conditions were checked by embedding a unit cell in a larger normalization sphere, and applying the boundary conditions at the surface of the larger sphere. The equilibrium composition, exterior gas density, and binding energy per particle in the unit cell were independent of the radius of the larger sphere, thereby establishing the validity of the method.

With the approximations specified above, it is straightforward to perform a self-consistent Hartree-Fock calculation for the nuclear wave functions in a unit cell, for a given cell radius and number of neutrons and protons. An initial guess is made for the single-particle potentials and effective masses and the radial wave functions and eigenvalues are calculated subject to the boundary conditions at the cell boundary. The lowest N neutron states and Z proton states are occupied, ρ and τ are calculated for neutrons and protons and new potentials and effective masses are obtained. The lowest N and Z states for the new potentials are then filled and the process is iterated until self-consistency is achieved Because of the multiplicity of local minima, it is necessary to try several dissimilar initial guesses for the potentials to assure that the absolute minimum-energy configuration has been obtained.

To determine the minimum binding energy per nucleon at a specified baryon density, one must search over the number of nucleons in the unit cell, thus specifying the cell radius, and the ratio of neutrons to protons, as well as the spatial distribution of nucleons described above The results of such a search for a preliminary version of this theory are shown as a contour plot in fig. 1. One notes several local minima generated by the energy fluctuations arising from the shell closures for 28, 40, and 50 protons, and in this case the absolute minimum occurs at $Z = 50$. In practice, the search is greatly expedited by considering only β-stable configurations At each iteration one computes $E(Z+1, N-1) \approx \mu_e+\mu_p-\mu_n+m_e+m_p-m_n$ and converts neutrons into protons plus electrons or vice-versa. The resulting β-stable self-consistent solution corresponds to the minimum along a line of constant $N+Z$ in fig. 1, and one has only to perform a single parameter search over the number of particles in a cell.

5. Results in the free neutron regime

The resulting ground state solutions for 11 densities are presented in table 3 and fig. 2. Since it is impossible to distinguish neutrons in nuclei from those in the gas, we adopt the unusual convention of labeling nuclei by the total number of nucleons in the unit cell. In fig. 2, the energies per particle for the ground state configurations, denoted by the crosses, are compared with the energies of a β-stable uniform gas of electrons, protons, and neutrons, thus demonstrating the significant gain in binding energy obtained by forming a Coulomb lattice of nuclei surrounded by a low-density neutron gas.

TABLE 3

Numerical results in the free neutron regime

n_b (cm^{-3})	N	Z	μ_n (MeV)	μ_p (MeV)	$\tilde{\rho}_G$ (fm^{-3})	$\tilde{x}$	$(E/A)-m_n$ (MeV)	$(E_{gas}/A)-m_n$ (MeV)
2.79×10^{35}	140	40	0.2	−26.8	4×10^{-5}	0.53	−1.425	0.436
4.00×10^{35}	160	40	0.3	−29.4	9.7×10^{-5}	0.53	−0.962	0.543
6.00×10^{35}	210	40	0.6	−29.5	2.6×10^{-4}	0.53	−0.462	0.692
8.79×10^{35}	280	40	1.0	−28.5	4.8×10^{-4}	0.53	−0.050	0.865
1.59×10^{36}	460	40	1.4	−29.4	1.2×10^{-3}	0.52	0.541	1.214
3.73×10^{36}	900	50	2.6	−33.6	3.0×10^{-3}	0.46	1.465	1.926
5.77×10^{36}	1050	50	3.3	−34.5	4.7×10^{-3}	0.45	1.996	2.408
8.91×10^{36}	1300	50	4.2	−35.8	7.8×10^{-3}	0.44	2.610	2.981
2.04×10^{37}	1750	50	6.5	−43.6	1.84×10^{-2}	0.35	4.097	4.422
4.75×10^{37}	1460	40	10.9	−54.0	4.36×10^{-2}	0.28	6.428	6.660
7.89×10^{37}	950	32	15.0	−68.3	7.37×10^{-2}	0.16	8.611	8.657

The quantities $\tilde{\rho}_G$ and $\tilde{x}$ are the approximate density of the exterior neutron gas and the approximate ratio of protons to neutrons in nuclei respectively, and cannot be defined uniquely due to density fluctuations All other quantities are defined in the text

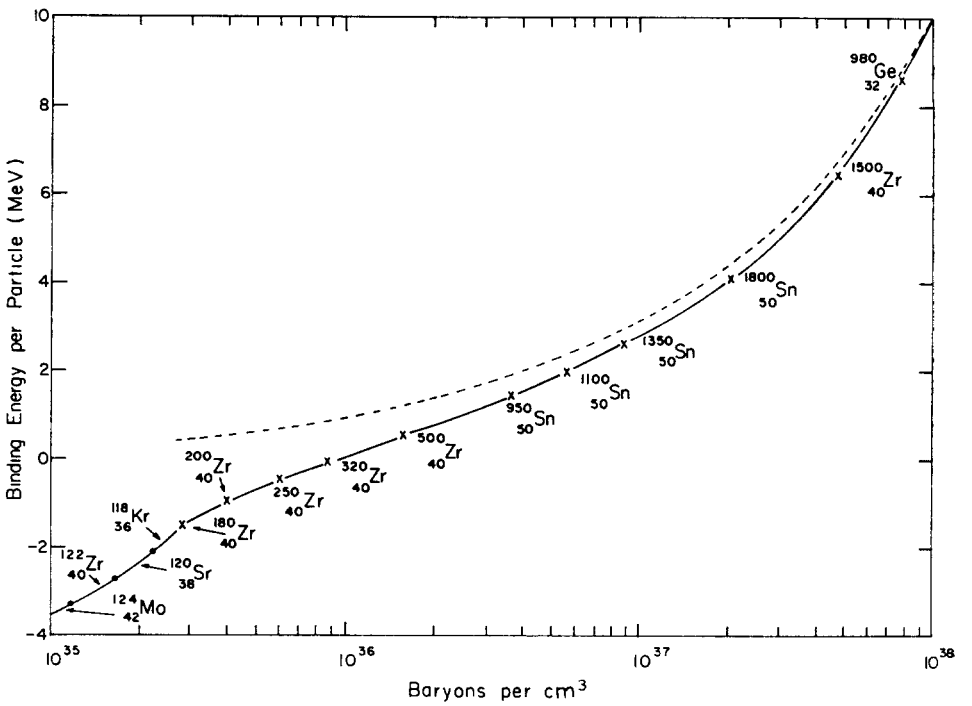


Fig 2 Energy per particle versus baryon density.

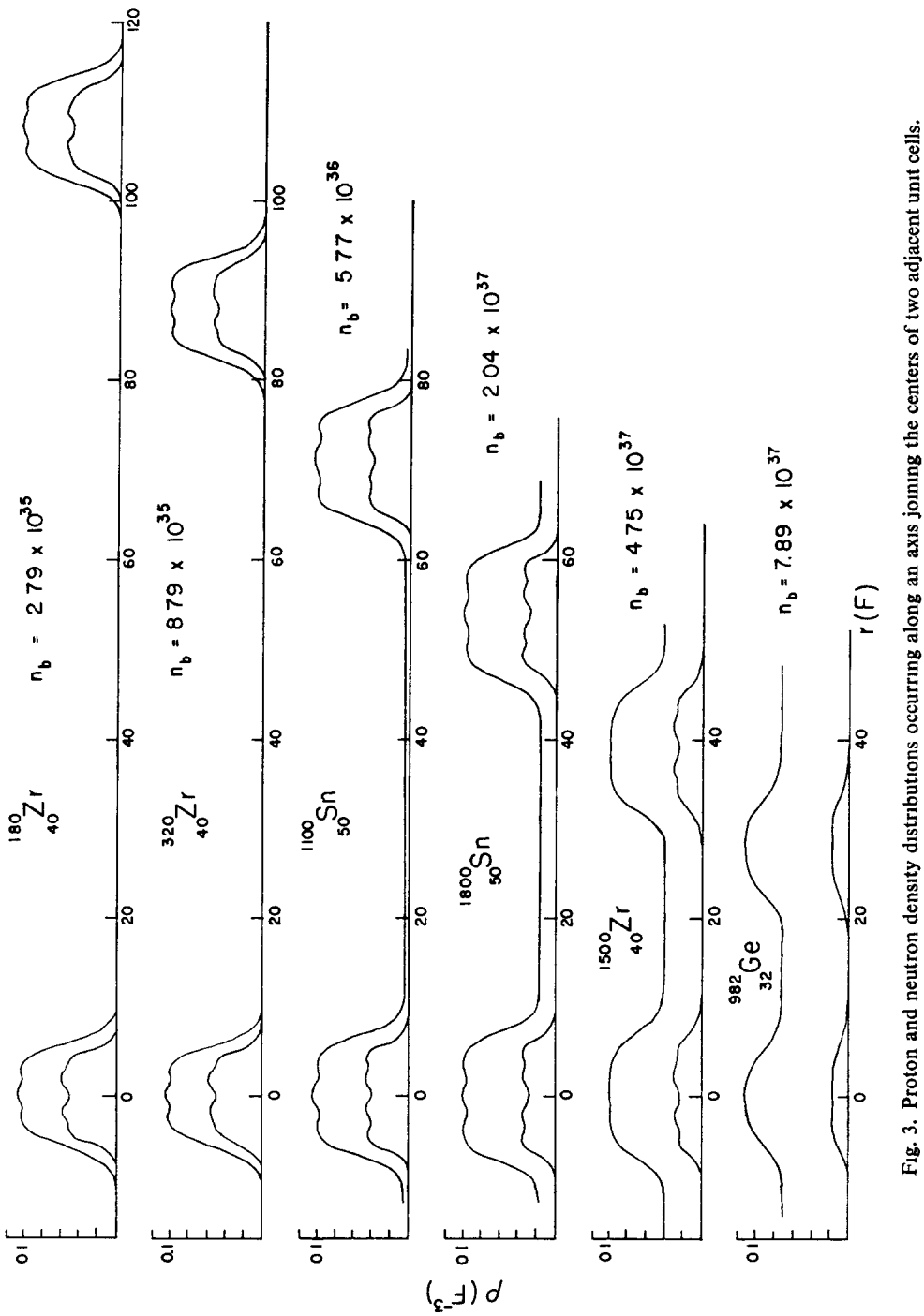


Fig. 3. Proton and neutron density distributions occurring along an axis joining the centers of two adjacent unit cells.

The spatial distribution of neutrons and protons at various densities is presented in figs. 3 and 4. Fig. 3 shows the densities obtained along a line joining the centers of two adjacent unit cells. As the baryon density increases, one observes a smooth,

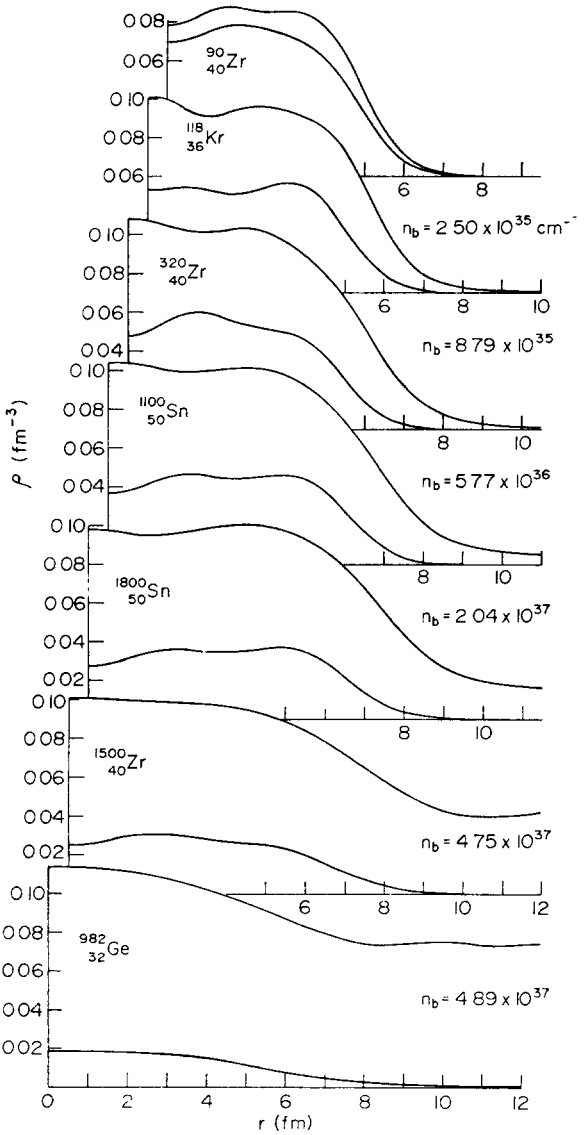


Fig. 4. Proton and neutron densities versus distance from the center of a unit cell

systematic progression of configurations. The cell radius decreases, the neutron gas density increases and the density of protons in nuclei decreases By the time one reaches $^{982}_{32}Ge$, the difference in energies of various local minima is sufficiently small

that it is not meaningful to proceed to higher density. The energy per baryon is already very close to the uniform gas result, and it seems quite reasonable to assume that the density distributions at higher density continue the systematic behavior and smoothly approach a uniform density

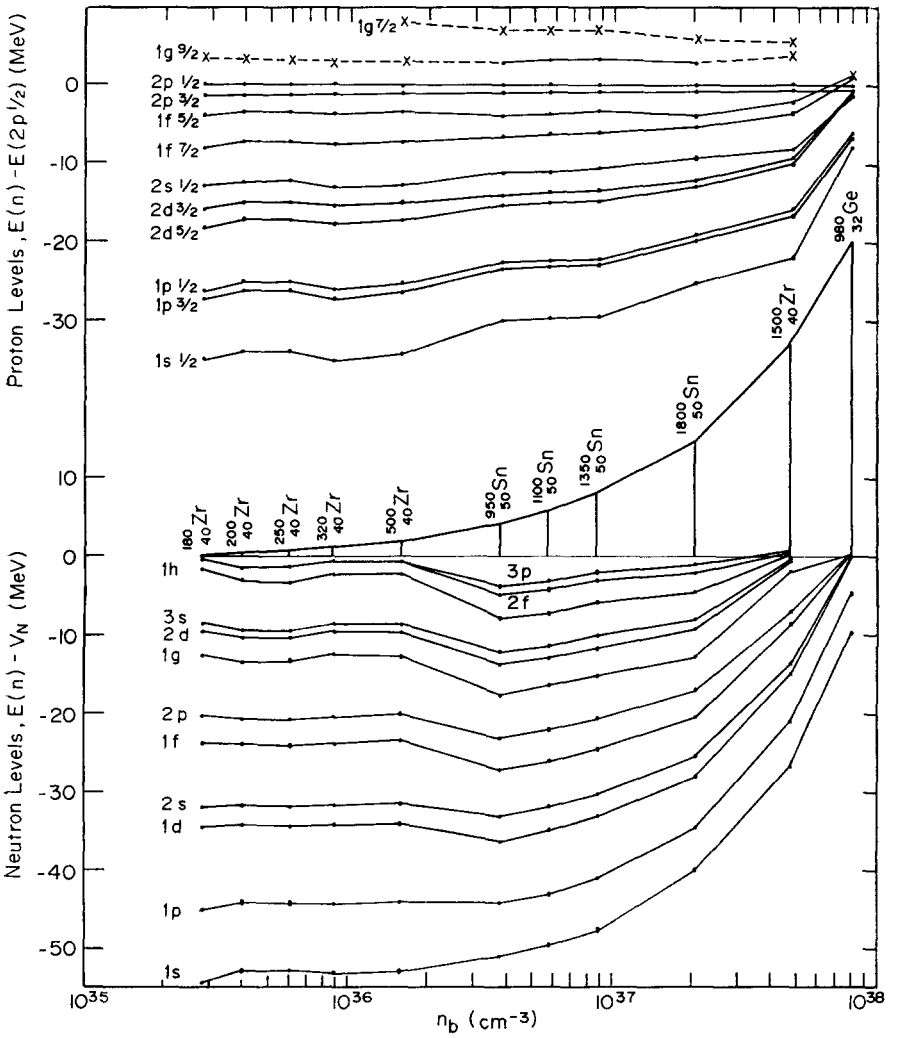


Fig 5. Single-particle spectrum of protons and neutrons

Unfortunately, we have not succeeded in formulating a tractable theory of the onset of fluctuations as the density of a uniform gas is lowered below some critical density, analogous to the Thomas-Fermi treatments by Baym *et al*, and Arponen [11]). The main impass has been the fact that the DME functional depends on both ρ and τ, and we have been unable to formulate the constraint between ρ and τ adequately

so as to express the energy associated with spatial fluctuations. Lacking an expression in terms of ρ and τ, it is legitimate to re-introduce the individual wave functions and write the RPA stability conditions in terms of $\delta^2 E(\delta\psi_i \delta\psi_j)$ using the DME functional, but this has not yet been carried out.

In fig. 4, the density distributions of the nuclei at the center of the unit cells are shown in greater detail. For comparison, $^{90}_{40}$Zr, a naturally occurring isotope, and $^{118}_{36}$Kr, the last pre-drip nucleus, are also presented. In addition to the systematic effects mentioned previously, one also observes a strong systematic increase of the nuclear surface thickness and the diminishing of neutron density fluctuations as the number of neutrons becomes sufficiently large and one approaches a statistical regime.

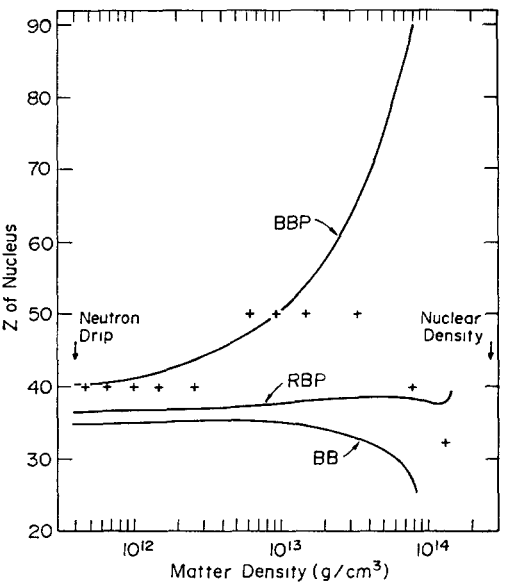


Fig. 6. Number of protons per nucleus, denoted by crosses, compared with previous predictions taken from ref [14])

One of the most striking features of figs 3 and 4 is the degree to which the nuclei in the free neutron regime resemble ordinary nuclei. This similarity is also manifested in the behavior of the single-particle energies, as shown in fig. 5. In order to eliminate irrelevant constant shifts in the absolute single-particle energies, relative energy spacings are plotted with the proton energies shown relative to the $2p_{\frac{1}{2}}$ state and the neutron energies shown relative to the continuum. The facts that the usual shell model level sequence is maintained throughout the free neutron regime and that the interior nuclear density does not deviate significantly from the density of ordinary nuclei strongly suggest that the DDHF theory should be just as reliable in this regime as in observable nuclei where it has been experimentally tested.

The compositions of the resulting nuclei in this theory are compared with the predictions of BBP [5]), Buchler and Barkat [13]) (BB) and Ravenhall, Bennett and Pethick (RBP) in fig 6. We have already argued that RBP is the most reliable of the three previous theories. The equilibrium conditions derived by BBP show that if the nuclear surface energy is expressed in the form BE $\sim W_{\mathrm{surf}} A^{\frac{2}{3}}$, then the equilibrium size of the nucleus is given by $W_{\mathrm{surf}} A^{\frac{2}{3}} = 2 E_{\mathrm{Coul}}$, where E_{Coul} is the total Coulomb energy per nucleus, including the lattice energy. However, in the spherical geometry of a Wigner-Seitz unit cell, in contrast to a plane surface, there are additional contributions

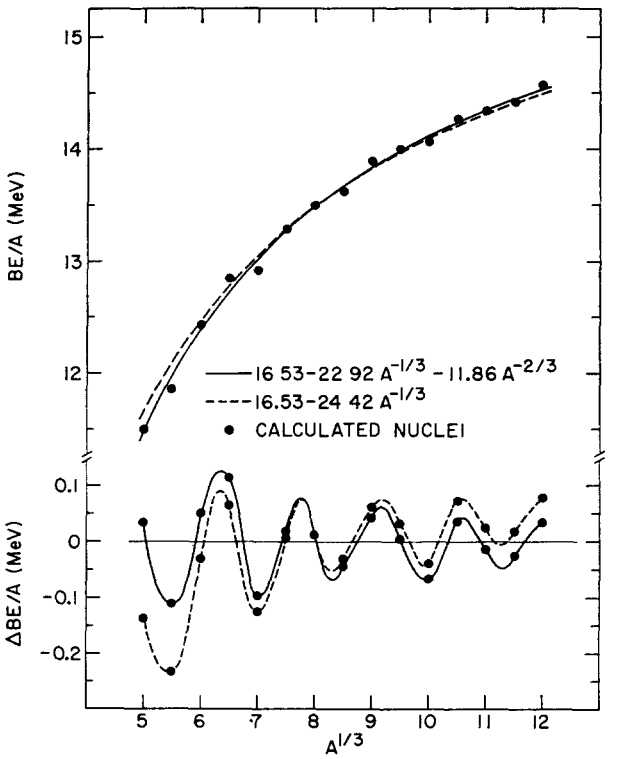


Fig. 7. Least-squares fit to the masses of large mirror nuclei to determine the curvature coefficient

to the surface energy proportional to $A^{\frac{2}{3}}$ arising from the curvature of the surface as well as a small pressure term which we will ignore for the present argument. The curvature term may be evaluated by calculating a sequence of unphysically large mirror nuclei with the Coulomb force turned off. The resulting binding energies are least-squares fit with a mass formula of the form $\mathrm{BE} = W_{\mathrm{vol}} A + W_{\mathrm{surf}} A^{\frac{2}{3}} + W_{\mathrm{curv}} A^{\frac{1}{3}}$ as shown in fig. 7. The resulting surface energy is rather different from the result of Myers and Swiatecki [24]) and will be discussed in detail in a subsequent paper dealing with the semi-empirical mass formula. If one repeats the argument of BBP with this

expression [25]), the new equilibrium condition is $(W_{surf}+2W_{curv}A^{-\frac{1}{3}})A^{\frac{2}{3}} = 2E_{Coul}$. Thus, substituting the values from fig. 7, although the curvature term in the binding energy is only about 11 % of the surface term for $A \approx 200$, it shifts the equilibrium size by 22 % Taking an average value of Z from RBP to be 37, this effect would be expected to shift the charge to 45. Because of shell effects, as demonstrated in fig. 1, the equilibrium point in our present theory is shifted away from 45, either to the shell closure at 40 or to the closure at 50, but it is significant that the average shift is accurately explained by this curvature effect

6. Equation of state

Given the total ground state energy per particle, including masses, E_T, as a function of baryon density, n_b, the pressure, adiabatic index and mass density are determined as follows:

$$P = n_b^2 \frac{\partial E_T}{\partial n_b}, \qquad \Gamma = \frac{n_b}{P}\frac{\partial P}{\partial n_b}, \qquad \rho = \frac{n_b E_T}{c^2}.$$

Since searches were carried out only at 11 discrete densities, it was necessary to fit the resulting energies shown in fig. 2 with a smooth function. The total energies per baryon, E_T, for both the ground state and uniform gas configurations were fit by the following function:

$$E_T = m_n + c_0 + \exp\{\sum_{I=1}^{7} c_I x^{(I-1)}\},$$

where E_T is in MeV, m_n is the neutron mass, $x = \ln(n_b \times 10^{-35})$, n_b is the baryon density in baryons per cm^3, and c_0–c_7 are tabulated in table 4.

TABLE 4
Coefficients for the equation of state

I	Ground State c_I	Uniform Gas c_I
0	−4.0	−4 0
1	2.8822899×10^{-1}	1.4821424
2	$5\ 9150523 \times 10^{-1}$	$-4\ 0373482 \times 10^{-2}$
3	9.0185940×10^{-2}	$6\ 0455728 \times 10^{-2}$
4	$-1.1025614 \times 10^{-1}$	$-1.5307639 \times 10^{-2}$
5	2.9377479×10^{-2}	3.4774416×10^{-3}
6	$-3.2618465 \times 10^{-3}$	$-4.3627154 \times 10^{-4}$
7	1.3543555×10^{-4}	$2\ 3383473 \times 10^{-5}$

The equation of state obtained by fitting the 11 ground state nuclear configurations in the free neutron regime is shown in fig. 8 by the solid line. At high density, it joins smoothly onto the BBP curve, denoted by the short dashed line. The equation of state for a uniform gas is indicated by the long dashed curve, and becomes indistinguishable from the ground state curve at intermediate densities. For comparison,

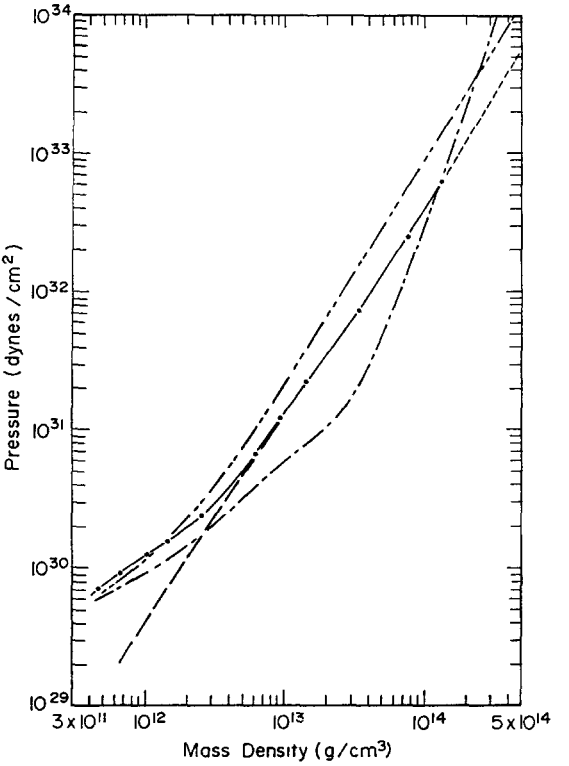


Fig. 8. Equation of state of this work (dots connected with solid lines) compared with that of uniform matter (long dashes), the BBP high-density results (short dashes), the Harrison-Wheeler equation of state (2 short dashes, 1 long) and the V_γ equation of state (short dash, long dash).

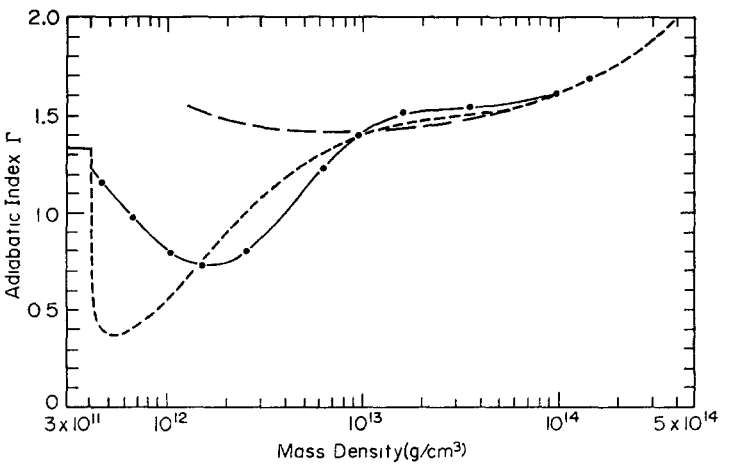


Fig 9. Adiabatic index compared with the BBP result of ref. [5])

the Harrison-Wheeler equation of state [26]) and the V_γ equation of state of Langer *et al.* [7]) are also plotted as taken from BBP.

The adiabatic index calculated with the present theory is shown in fig. 9 by the solid line. One should note that taking the second derivative of the curve fit to the eleven calculated energies strongly amplifies noise in the fit, and therefore undue significance should not be attached to the fine details of this curve. For comparison, the adiabatic index obtained for a uniform gas is shown by the long dashes and the BBP result is shown by the short dashes.

7. Conclusion

We have attempted to show in this work that it is possible to construct a reliable theory of a nucleon many-body system derived from the two-body nucleon-nucleon interaction. The relevant two-body correlations are incorporated in a two-body effective interaction, and the energy density is expressed as an extremely simple functional of the density and kinetic energy density via the density matrix expansion. This theory yields excellent agreement with experimentally observable properties of finite nuclei and should provide a reliable extrapolation to the nucleus-like clusters occurring in the free neutron regime of dense matter.

The primary uncertainty affecting the equation of state we have calculated in the free neutron regime is the small uncertainty in the function specifying the uniform matter potential energy for a small ratio of protons to neutrons. The adjustment for the higher-order corrections, the ambiguity in off-energy-shell behavior of the nuclear force, and the omission of pairing correlations, all contribute to the uncertainty in the energy of a neutron gas. Fortunately, throughout most of the free neutron regime, the nuclear clusters occurring in the center of the unit cell are insensitive to small changes in the energy functional for a neutron gas, so that the composition and energy of the cluster is essentially unaffected. Since most of the baryons in the cell are neutrons in the gas, their energy change may be computed directly from the change in the neutron gas energy and thus the change in the total energy and therefore the equation of state may be obtained straightforwardly from the data in table 3 without repeating the lengthy self-consistent calculations reported in this work.

Finally, the substantial agreement between the final equation of state obtained in this work and that of BBP suggests that calculations of Baym *et al.* [3]) based on the BBP results should be essentially unchanged by this present work.

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